

Dimensionality Reduction

Seeing Complex Data in Simple Ways
By Muhammad Nanda S. / Data Scientist



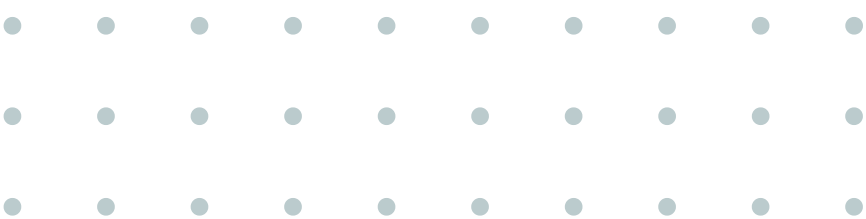
Session Objectives

01. Understand dimensionality reduction and **why it's needed**

02. Learn Principal Component Analysis (**PCA**) as a **technique**

03. Explore **practical applications** of PCA

04. Interpret PCA results

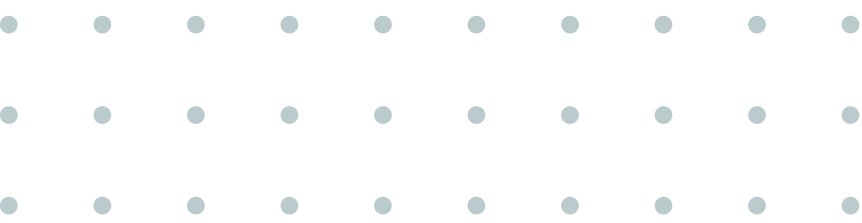


Why Reduce Dimensions?

When data often has too many features

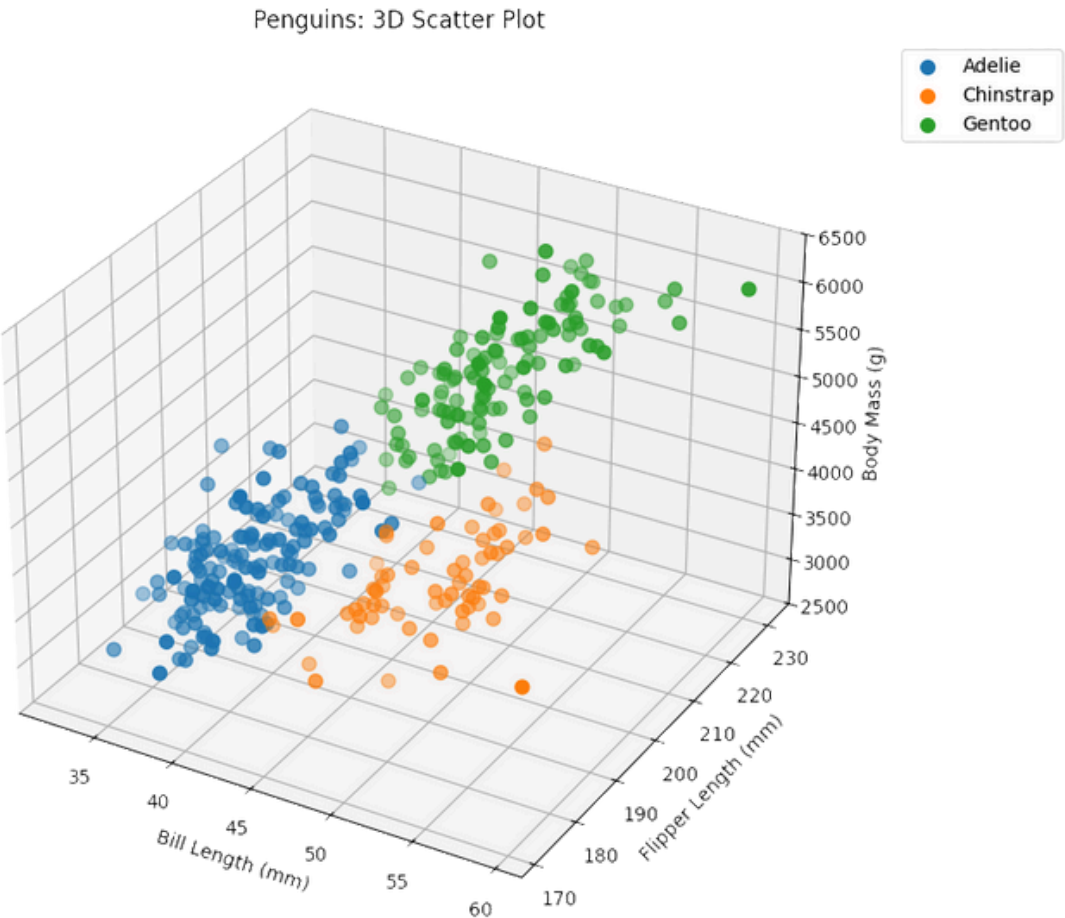
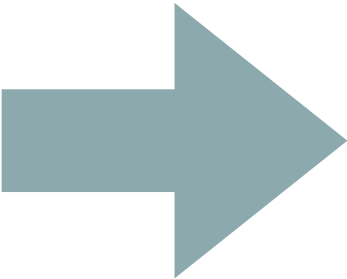
- Hard to visualize beyond 3D
- Some features are repetitive or useless
- More dimensions can actually hurt model performance (**curse of dimensionality**)

▲ species A factor denoting penguin species (Adélie, Chinstrap and Gentoo)	▲ bill_length_mm A number denoting bill length (millimeters)	▲ flipper_length_mm An integer denoting flipper length (millimeters)	▲ body_mass_g	▲ sex An integer denoting body mass (grams)
Adelie44%	41.12%	1906%	38003%	male49%
Gentoo36%	45.22%	1955%	37003%	female48%
Other (68)20%	Other (331)96%	Other (305)89%	Other (321)93%	Other (11)3%
Adelie	39.1	181	3750	male
Adelie	39.5	186	3800	female
Adelie	40.3	195	3250	female
Adelie	NA	NA	NA	NA
Adelie	36.7	193	3450	female
Adelie	39.3	190	3650	male
Adelie	38.9	181	3625	female
Adelie	39.2	195	4675	male



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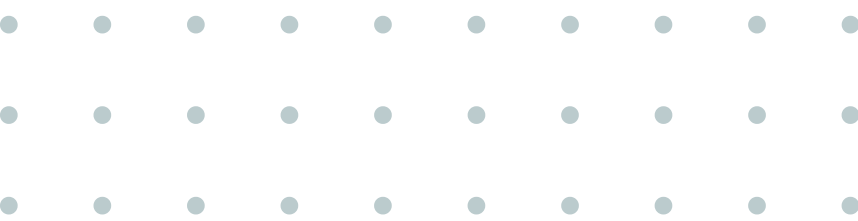
Δ species		Δ bill_length_mm		Δ flipper_length_mm		Δ body_mass_g	
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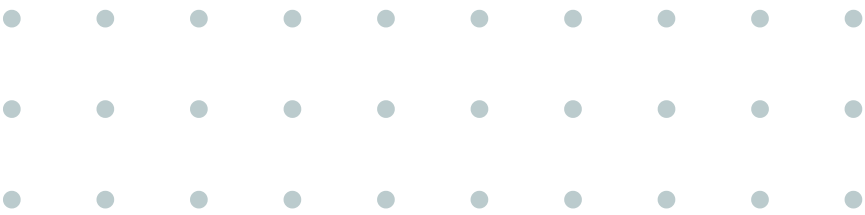
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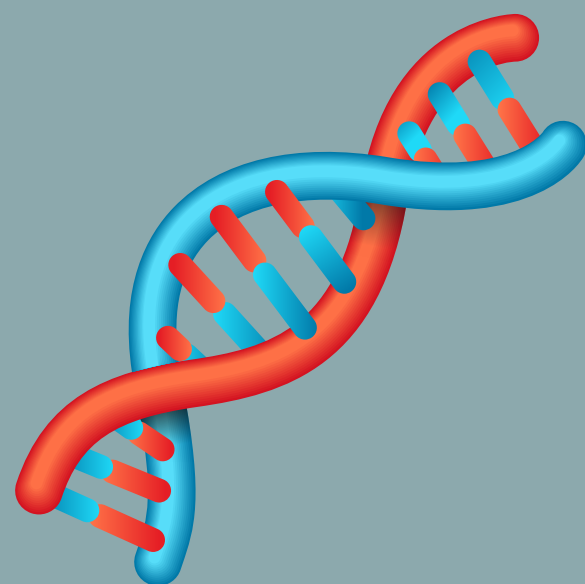


Practical Use Cases of PCA (beyond visualization)

- **Faster Machine Learning**
 - Fewer features → less overfitting, faster training
- **Hidden Patterns** (Feature Engineering)
 - New composite features reveal underlying behavior
- **Data Compression**
 - Store less, keep the essence of the data
- **Pre-step** for Other Algorithms
 - Used before clustering (like k-means) to make clusters clearer.



Where do we use PCA?



Genetics

reduce thousands of genes to
a few components



Face Recognition

compress pixels into key
features ("eigenfaces")

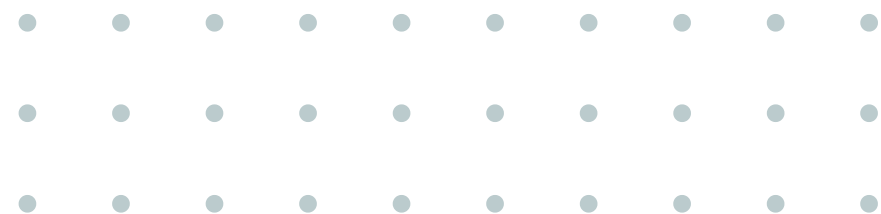


EDA

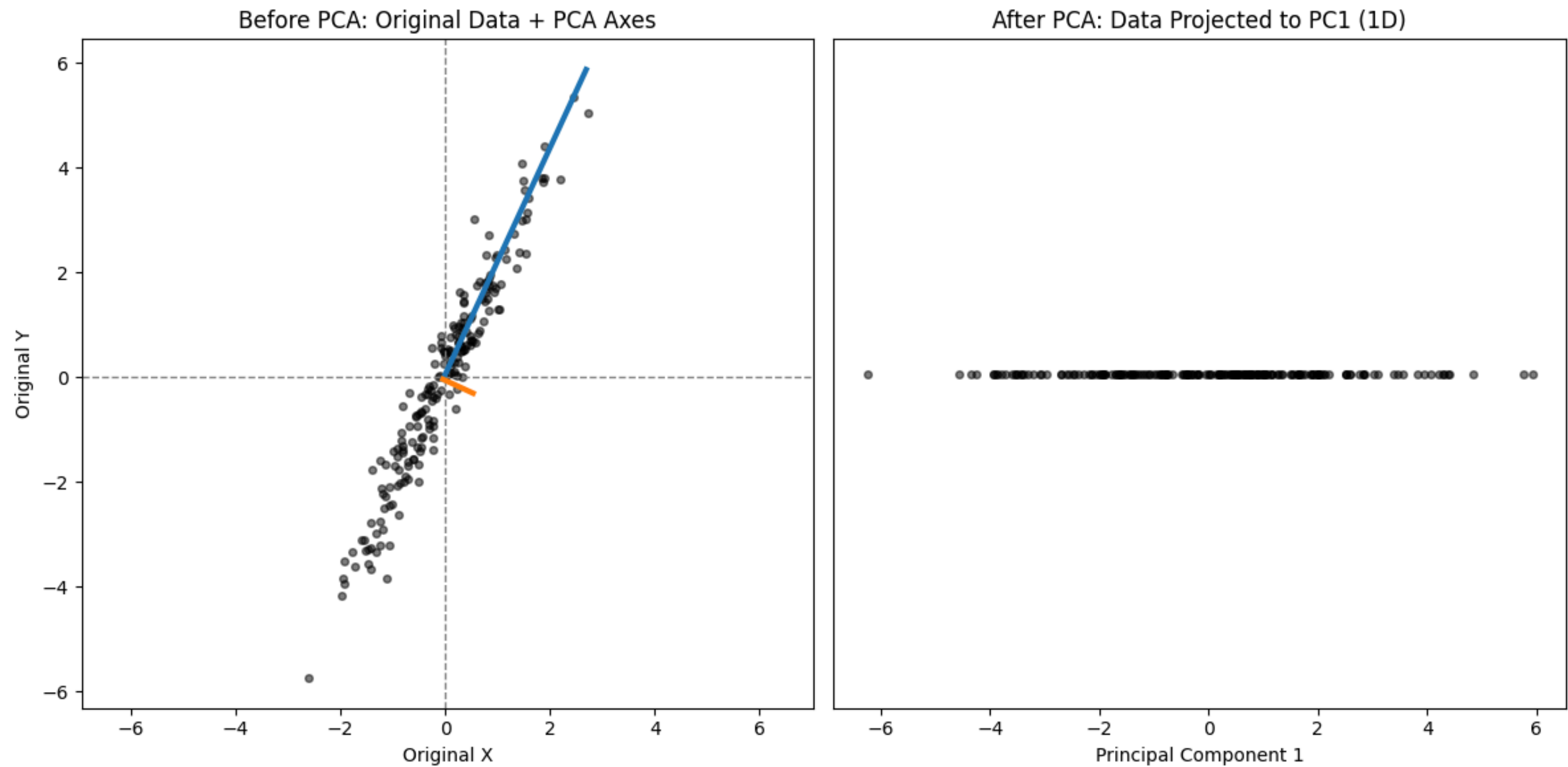
reveal hidden structure in
complex data

What is PCA?

- A statistical technique to **simplify complex data**
- Transforms data into a **new coordinate system**
- First component: direction with the most variance, the most importance
- Next components: capture remaining variance, orthogonal to previous



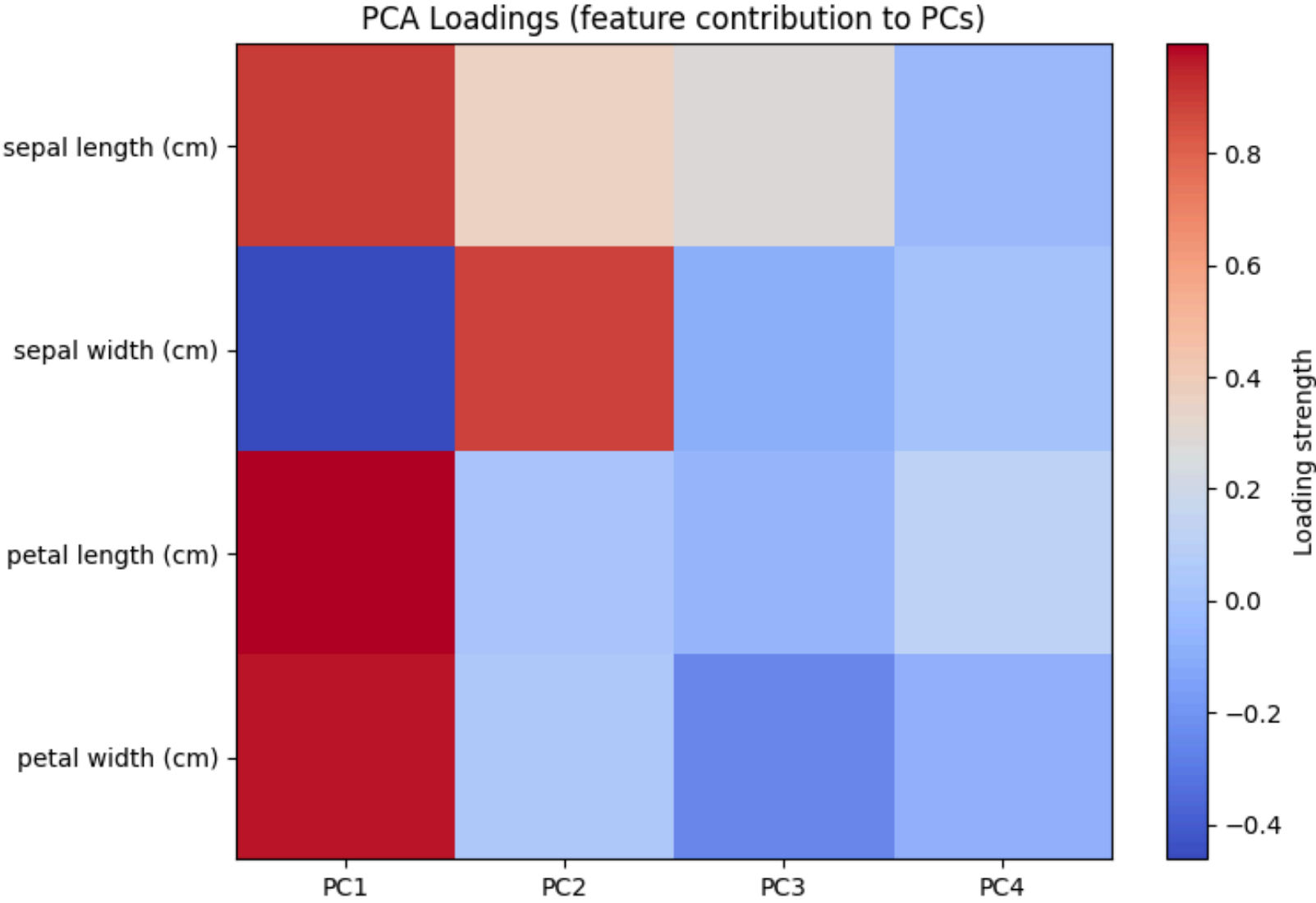
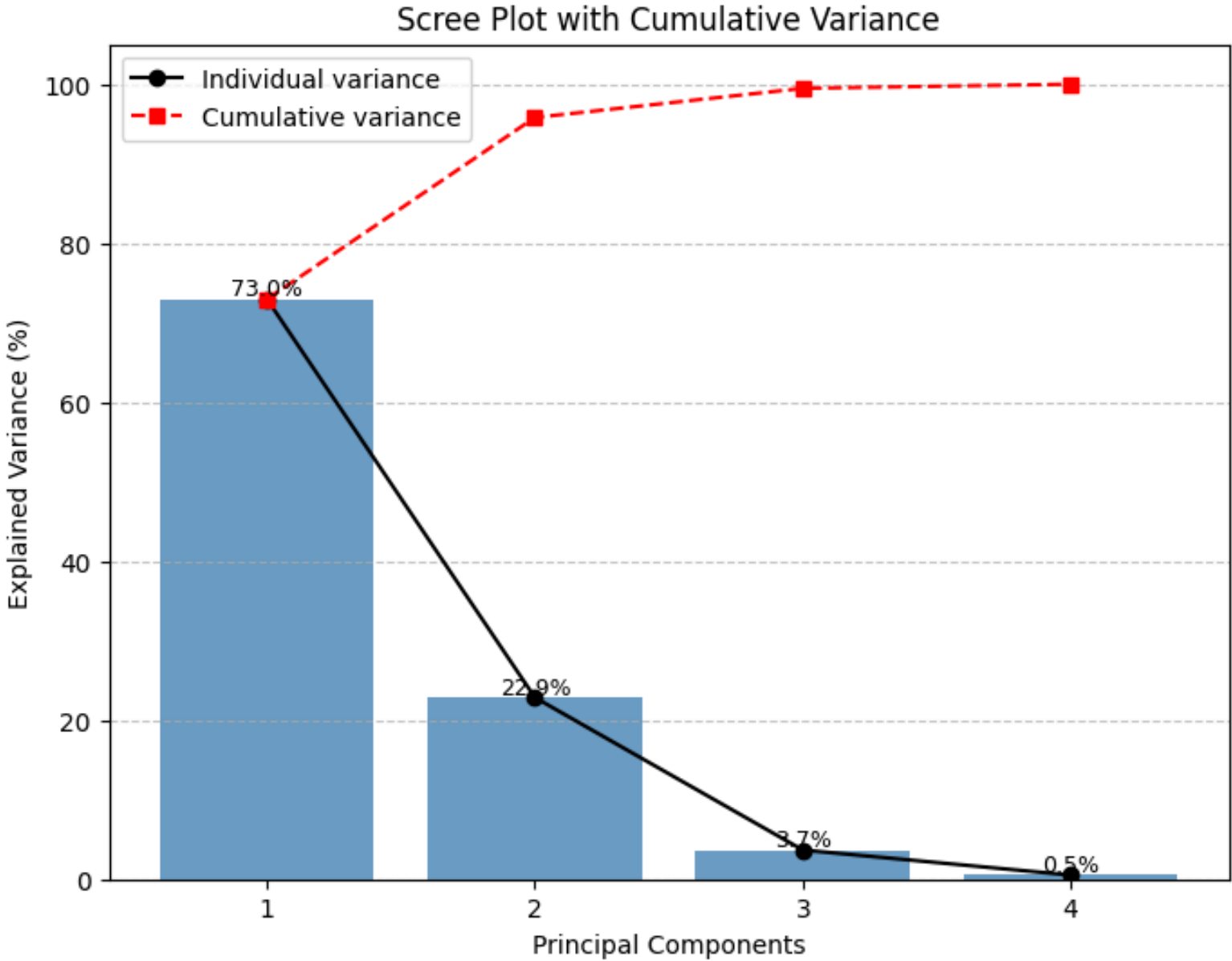
PCA Example



PCA is just **projection onto new basis vectors** (eigenvectors of the covariance matrix)
Steps: Data → Standardize → Covariance → Eigenvectors → Select PCs → Reduced Data



Reading and Using PCA Results



Strength & Limitation

✓ Strengths

- Reduces noise
- Enables easier visualization
- Speeds up computation

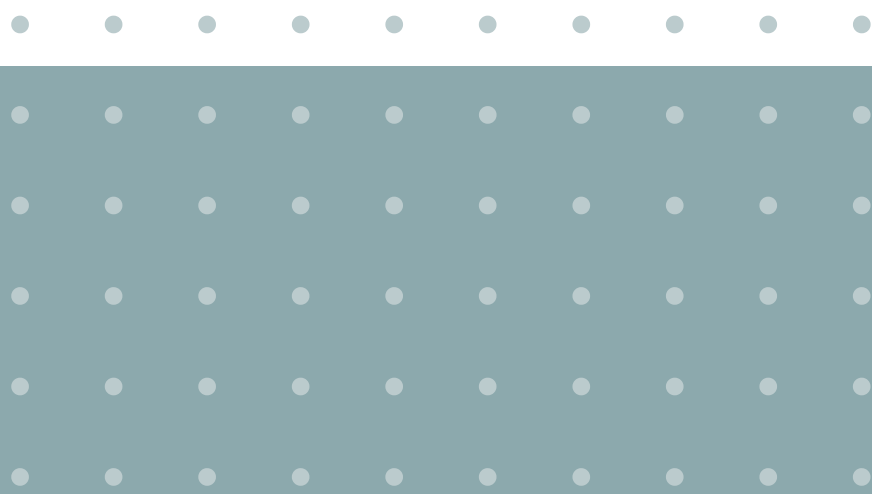
⚠ Limitations

- Assumes linearity
- Interpretation can be tricky
- Some information is lost

Trade-off:

- Fewer PCs → simpler model, but risk of losing details
- More PCs → retain more information, but less reduction





Let's Discuss!

Do you have any question?

